Akshay Dahiya

SENIOR PROTOCOL ENGINEER

Email | Portfolio | GitHub | LinkedIn | Twitter | Telegram | India (UTC +5:30)

PROFESSIONAL SUMMARY

Senior protocol engineer with 15+ years of software development experience, including 6+ years in blockchain and web3 infrastructure. Core expertise spans prediction markets, smart contract engineering, tokenomics design, gasless relayers, decentralized data systems, and high-throughput API architecture. Proven track record building production DeFi protocols and live prediction-market infrastructure.

Seeking Senior Protocol Engineer roles focused on prediction markets, DeFi infrastructure, data indexing, or protocol development.

KEY ACHIEVEMENTS

- **Live prediction-market systems** — Polymarket 5m BTC up/down stack: fee-accurate backtests, CLOB execution, fill forensics
- **~800 ETH raised + 5,300+ SBT mints** via Snapshotter Node contracts
- **1M+ gasless transactions** sponsored through EIP-712 relayers
- **Full protocol lifecycle** — led engineering from PoC → 3 testnets → Mainnet V1 → V2
- **Generative Prediction Markets (GPM)** — auto-generated 5-minute Uniswap markets resolved from snapshotter consensus, not external oracles
- **70+ audited smart contracts** deployed across EVM chains including Play-to-Earn, MEV-resistant vaults, and cross-chain bridges
- **3M+ daily API requests** served; reduced latency by 80% through optimization
- **2 ACM publications** featured in Forbes; built counter-plagiarism AI (2M+ exam pairs) still in production at Georgia Tech
- **1,000+ students mentored** and **2,000+ projects reviewed** across Udacity, 4-year GSoC Admin, and 5+ ETHGlobal hackathons

TECHNICAL SKILLS

Languages	Python (14 yrs), JavaScript/Node.js (5 yrs), Solidity (6 yrs), Swift/SwiftUI
Prediction Markets	CLOB Execution, Conditional Tokens/FPMM, Oracle Resolution, Market Microstructure
Protocol & Tokenomics	Protocol Design, Game Theory, Consensus, Staking/Slashing, Tax Curves, Airdrops, Sybil Resistance, Incentivized Testnets, Emission Schedules, Delegation, Incentive Alignment
Smart Contracts	Upgradeable Proxies, EIP-712, Gasless Relayers, VRF, Oracles, ERC20/721/1155, Audited Code, SBTs
Decentralized Systems	Snapshotter Networks, Sequencers, IPFS/Data Availability, Prediction Markets, Points Systems, Agentic AI
Backend & DevOps	Linux/Shell, Docker, K8s, CI/CD, AWS, Redis, PostgreSQL, High-Throughput APIs
Web3 Tooling	Hardhat, Foundry, Ethers.js, Web3.py, IPFS, Data Indexing, Chainlink, Polymarket CLOB
AI/ML	PyTorch, TensorFlow, LLMs, Agentic AI, Reinforcement Learning, Computer Vision
Frontend	React, Next.js, SwiftUI

PROFESSIONAL EXPERIENCE

Founding Engineer

Jan 2026 - Present

STEALTH STARTUP — *Remote*

- Built live trading infrastructure for Polymarket 5-minute BTC up/down markets — Binance/Chainlink oracle lag, CLOB execution, and on-chain settlement/redemption
- Shipped a fee-accurate backtest engine (Nautilus/Polars) plus live forensics for fill quality, kill switches, and bankroll risk limits
- Built SwiftUI iOS apps (Core Data, HealthKit) and applied modern LLMs and agentic tooling across research and production workflows

Lead Protocol Engineer & DeFi Researcher

Dec 2022 - Jan 2026

Powerloom Protocol — *Remote*

- Co-designed core protocol evolution from V0 → V1 → V2 → DSV architecture
- Engineered Snapshotter node registry contracts, securing **5,300+ nodes** and **~800 ETH** raised
- Designed **Powerloom protocol contracts**, staking modules, and airdrop mechanics
- Built high-throughput **EIP-712 relayers** ensuring gasless txs for **1M+ interactions**
- Designed and shipped **Generative Prediction Markets (GPM)**: auto-generated 5-minute Yes/No Uniswap price markets whose strikes and outcomes came from snapshotter consensus — no operators, no external oracles
- Built the GPM on-chain lifecycle: pool/strike selection, gasless betting via EIP-712 relayers, $POP collateral through **Onchain Points**, automatic resolution, and Farcaster Frames
- Led engineering for **3 incentivized testnets** with partners like CoinList, Polygon zkEVM, and Socket
- Single-handedly shipped **BDS (Blockchain Data Services)**: decentralized real-time Uniswap data with IPFS proofs

Blockchain Developer

Dec 2020 - Nov 2022

Independent Contractor — *Remote*

- Delivered **50+ audited smart contracts** across EVM chains, prioritizing gas optimization
- Architected complex tokens (ERC-20/721/1155) with deflationary and dynamic reward logic
- Built DeFi protocols including MEV-resistant vaults and automated referral systems
- Developed automated trading infrastructure using Web3.py and Solidity for NFTs and ERC20 tokens
- Engineered Play-to-Earn ecosystems integrated with Chainlink VRF and cross-chain bridges

Senior Software Engineer

Mar 2020 - Dec 2022

Hypersonix Inc. — *Remote*

- Engineered high-availability **API Gateway** scaling to **3M+ daily requests** for enterprise data
- Optimizing API latency by **80%** using Brotli compression and query tuning
- Implemented enterprise auth with Auth0 (SSO, OAuth 2.0) to secure sensitive endpoints
- Deployed comprehensive ELK observability stack for real-time monitoring

Graduate Programmer

Jun 2018 - Nov 2018

Fidelity International — *Bangalore, India*

- Built real-time survey platform using WebSockets, Django, Redis, and D3.js
- Developed React-based analytical tools for behavioral finance portfolio managers

Technical Mentor & Project Reviewer

Feb 2017 - Mar 2020

Udacity — *Remote*

- Reviewed **2,000+ technical projects** across Full Stack and AI nanodegrees
- Mentored **500+ students** globally through code reviews and debugging sessions

Open Source Contributor & Admin

Feb 2018 - Aug 2021

Google Summer of Code — Hydra Ecosystem — *Remote*

- Core contributor to Hydra Ecosystem — framework for smart, self-describing Web APIs
- Served as GSoC Admin/Mentor (2018–2021), guiding **10+ contributors** on open-source tooling

EDUCATION

Master of Science in Computer Science — Machine Learning Specialization

2019 - 2021

Georgia Institute of Technology — *Atlanta, GA*

- Research: Built **Examinator** (ML plagiarism detection) and **JackMarker**; published 2 ACM papers
- Coursework: Machine Learning, Reinforcement Learning, Computer Vision, AI, Graduate Algorithms

Nanodegree Certifications

2016 - 2020

UDACITY

Online

- **Deep Learning & Computer Vision (2020):** Neural networks with PyTorch — CNNs, RNNs, LSTMs, GANs
- **React (2017):** 3 full-stack apps with Redux — MyReads, Would You Rather, Mobile Flashcards
- **Full Stack (2016):** PostgreSQL, RESTful APIs, OAuth, AWS deployment

Open Courseware Foundations

2015 - 2019

MIT OCW / EDX

Online

- **Algorithms (MIT 6.006 / Stanford CS161):** Heaps, BSTs, graph algorithms, dynamic programming, NP-completeness
- **MIT 6.042J:** Discrete math — proofs, graph theory, probability
- **Berkeley CS188:** AI fundamentals — search, MDPs, reinforcement learning, Q-learning

PUBLICATIONS

Research featured in Forbes **and** Times Higher Education. Systems remain in production at Georgia Tech.

1. Dahiya, A., et al. **JackMarker with GitDown: A Framework to Counter Plagiarism at Scale**
ACM Learning @ Scale 2021. DOI: 10.1145/3430895.3460982

2. Apoorv, R., Dahiya, A., et al. **Examinator: A Plagiarism Detection Tool for Take-Home Exams**
ACM Learning @ Scale 2020. DOI: 10.1145/3386527.3406723

OPEN SOURCE WORK

Powerloom Protocol — Core protocol infrastructure for decentralized data markets: github.com/powerloom

Onchain Points & GPM — Points ecosystem and Generative Prediction Markets: github.com/Onchainpoints

Hydra Ecosystem — Linked Data framework for semantic Web APIs (GSoC 2018-2021): github.com/HTTP-APIs